



## Abstract

KENO is a popular lottery game that is played all over the US, and is currently played in 25 states in the U.S. Each state lottery has their own set of rules and prize structure. This led to a simple question: which state's game will you win the most amount of money? To answer this, the theoretical expected value and simulated expected value of the return were calculated utilizing MATLAB and R. The overall highest expected value is from Virginia's One Spot KENO game with a return of 75 cents per dollar played.

## Methods/Simulations

- Prize data was gathered from each state lottery's official website.
- The probability for a given match was calculated using the hypergeometric distribution density formula (Eq. 2), and the Expected Value associated with each match was calculated using Eq. 1.
- The theoretical calculations were cross-checked empirically with simulations in MATLAB and R.
- Figure (1). showcases side-by-side the simulation results and respective theoretical calculations for each spot in Missouri Keno.

| Spot Game | Simulated Expected Value | Theoretical Expected Value |
|-----------|--------------------------|----------------------------|
| 1         | 0.49995                  | 0.5                        |
| 2         | 0.60119                  | 0.60127                    |
| 3         | 0.62464                  | 0.62439                    |
| 4         | 0.61267                  | 0.61268                    |
| 5         | 0.62247                  | 0.62254                    |
| 6         | 0.60028                  | 0.60029                    |
| 7         | 0.60281                  | 0.60417                    |
| 8         | 0.58076                  | 0.58203                    |
| 9         | 0.58486                  | 0.58497                    |
| 10        | 0.60168                  | 0.60155                    |

Figure (1). Missouri Simulated (500,000,000 Games) vs Theoretical Expected Value

## Results

- Maximum expected value ranged from 0.502 (50 cents) to 0.75 (75 cents), as illustrated by Figure (2).
- Virginia 1-Spot KENO has the highest expected value out of all states, being 0.75, depicted in Figure (2).
- Average expected value in Virginia is about 0.72, which exceeds the maximum expected value of all other states.

### Overall Best Spot:

- The 7-spot game has the highest average expected value across all states.
- The 5 and 2 spot games most frequently produce the highest expected value, represented by Figure (3).

## Conclusion

To maximize returns, the 1-spot game in Virginia provides the highest expected value among all states. For a more general strategy, 2-spot and 5-spot games offer the highest returns for a combined 56% of states. The 7-spot game has the highest average return across all states. This makes it the best choice if you don't know the exact best spot for your state.

## References

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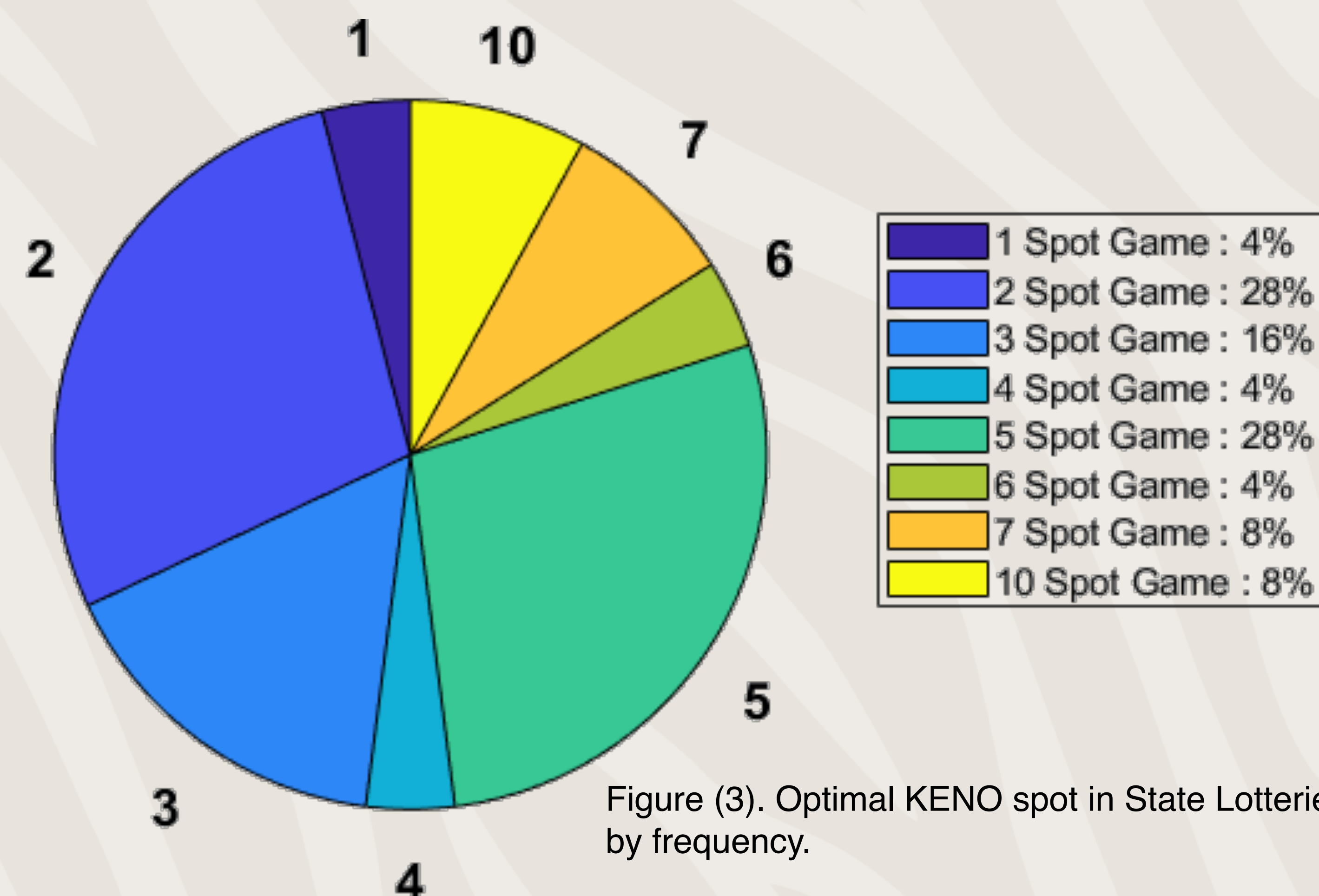


Figure (3). Optimal KENO spot in State Lotteries by frequency.

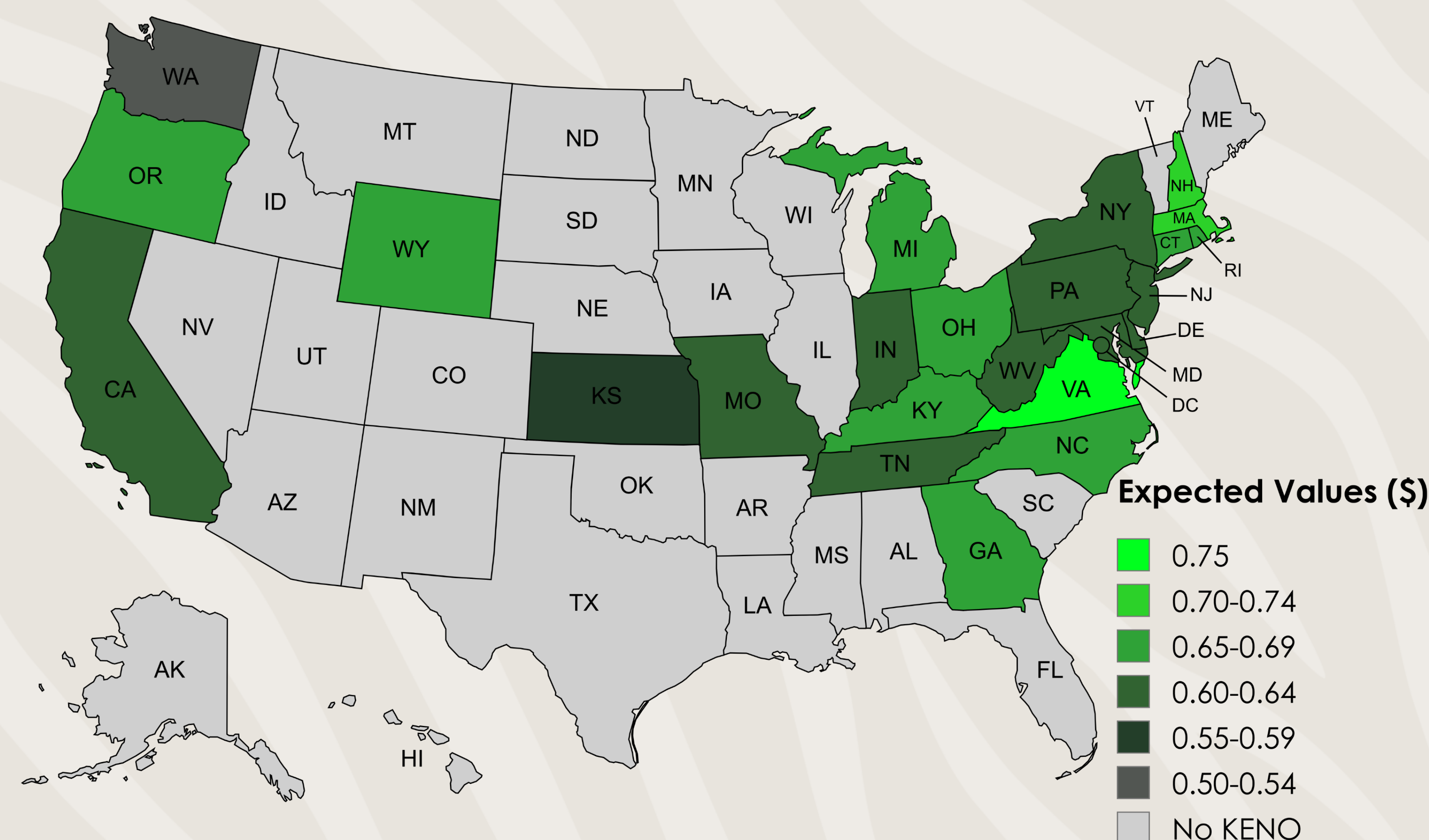


Figure (2). Highest Expected Value for all State Lottery KENO

## Steps to Play Keno

- The player first decides the number of spots they would like to play (majority of states offer 1 to 10 spot games).
- The player must choose their spots from the 1-80 available.
- The house (generally a computer) selects 20 numbers at random as "winning" numbers for that cycle.
- Prizes are awarded based on the number of player numbers that were also selected by the computer. These are denoted as "matches".
- In general, matching more spots nets the player a larger prize, which varies from state to state.

$$Expected\ Value = \sum_{i=1}^n P_i \times V_i$$

$P_i$  = Probability of Outcome  
 $V_i$  = Value of Prize

Equation (1). Expected Value Formula

$$P(X = k) = \frac{\binom{n}{k} \binom{80-n}{20-k}}{\binom{80}{20}}$$

$n$  = Number of Spots Played  
 $k$  = Number of Matches

Equation (2). Hypergeometric Distribution